

Model diagnostics

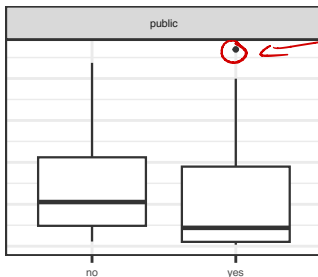
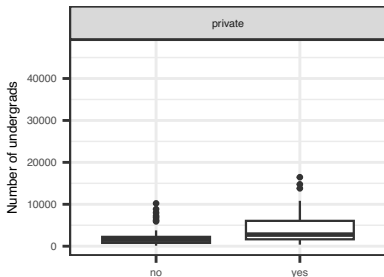
Ciaran Evans

Warmup

Work on the warmup activity (handout), then we will discuss as a class.

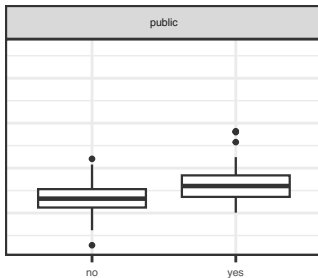
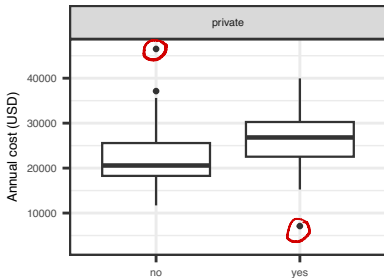
Warmup

effect of # undergrads differs for public vs. private
⇒ interaction: # undergrads × School type



doesn't
follow
general
trend

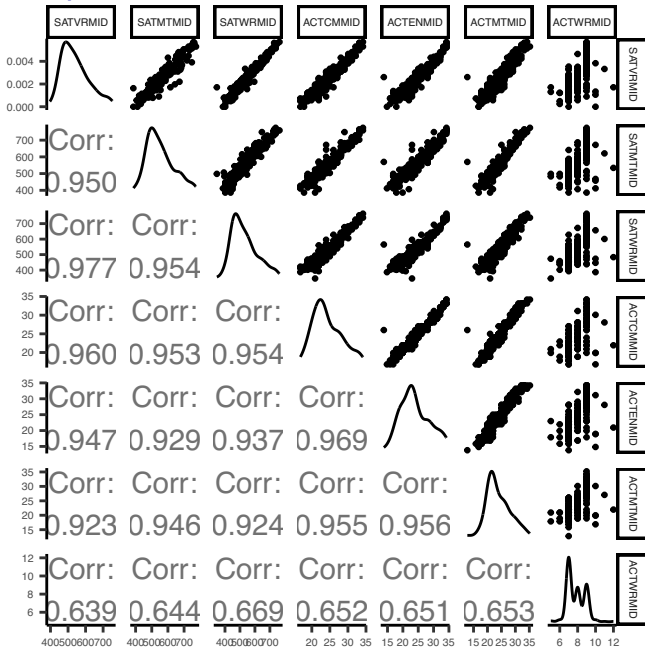
At least 70% of students repaying loans?



At least 70% of students repaying loans?

Warmup

Strong correlations!



why are correlated columns a problem?
 $n \times (k+1)$

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1k} \\ \vdots & x_{21} & x_{22} & \dots & x_{2k} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nk} \end{bmatrix} \in \mathbb{R}$$

Suppose two columns are perfectly correlated

$$\Rightarrow \text{rank } X < k+1$$

IRLS: $\hat{\beta} = (X^T W X)^{-1} X^T W z$

Also: $\hat{\beta} \approx N(\beta, (X^T W X)^{-1})$

$\Rightarrow$ need $(X^T W X)^{-1}$

But: $X^T W X \in \mathbb{R}^{(k+1) \times (k+1)}$

. If $\text{rank } X < k+1$, $\text{rank } X^T W X \leq \text{rank } X$
 $X^T W X$ is not invertible!

. If correlation is high, $X^T W X$ is close to singular

$\Rightarrow$ inverse unstable

$\Rightarrow$ high variability in $\hat{\beta}$

Outliers and multicollinearity

Outliers: Observations which don't follow the same trend as the rest of the data.

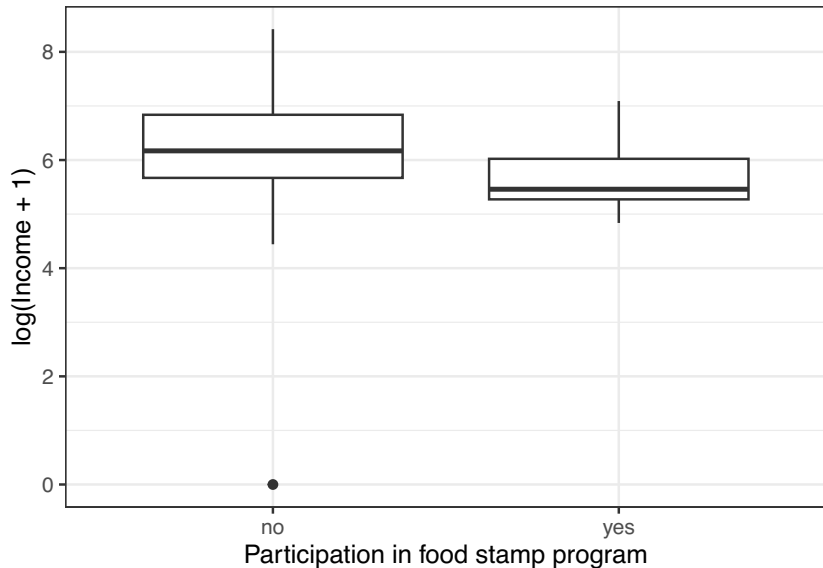
- examine any unusual points in EDA
- leverage (potential for influence)
 { Cook's distance (after fitting a model)

Multicollinearity: (linear) relationship between 2 or more explanatory variables.

- look at pairs plots, { correlation between explanatory variables
- variance inflation factor (VIF)

Both can cause problems for estimating parameters in a regression model.

Example: outliers



Example: outliers

With potential outlier:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.061	1.663	1.239	0.215
log(income + 1)	-0.617	0.278	-2.224	0.026

Without potential outlier:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	5.961	2.276	2.619	0.009
log(income + 1)	-1.274	0.391	-3.257	0.001

Cook's distance

Cook's distance for i^{th} observation:

$$C_i = \frac{1}{k+1} (\hat{\beta} - \hat{\beta}_{(i)})^T (\hat{V}_{cr}(\hat{\beta}))^{-1} (\hat{\beta} - \hat{\beta}_{(i)})$$

parameters

estimate w/ full data

estimate when i^{th} obs. is removed

Motivation:

$$\hat{\beta} \approx N(\beta, \hat{V})$$

$$(e.g. \hat{V} = (X^T W X)^{-1})$$

$$\Rightarrow (\hat{\beta} - \beta)^T \hat{V}^{-1} (\hat{\beta} - \beta) \approx \chi^2_{k+1}$$

$$\Rightarrow 1 - \alpha \text{ confidence region: } \left\{ \beta : (\hat{\beta} - \beta)^T \hat{V}^{-1} (\hat{\beta} - \beta) \leq \underbrace{\chi^2_{k+1, \alpha}}_{\text{upper } \alpha \text{ quantile of } \chi^2_{k+1}} \right\}$$

Common guideline: concerned if $C_i > 1$

$$\chi^2_{k+1, 0.5} \approx k+1 \Rightarrow \frac{1}{k+1} \chi^2_{k+1, 0.5} \approx 1$$

$C_i > 1 \Leftrightarrow \hat{\beta}_{(i)} \underline{\text{not}}$ in 50% confidence region for β

Cook's distance

Shortcut formula:

$$C_i \approx \frac{(y_i - \hat{p}_i)^2}{(n+1) \hat{p}_i (1 - \hat{p}_i) (1 - h_i)} \cdot \frac{h_i}{(1 - h_i)}$$

how unusual is observation i , potential to influence

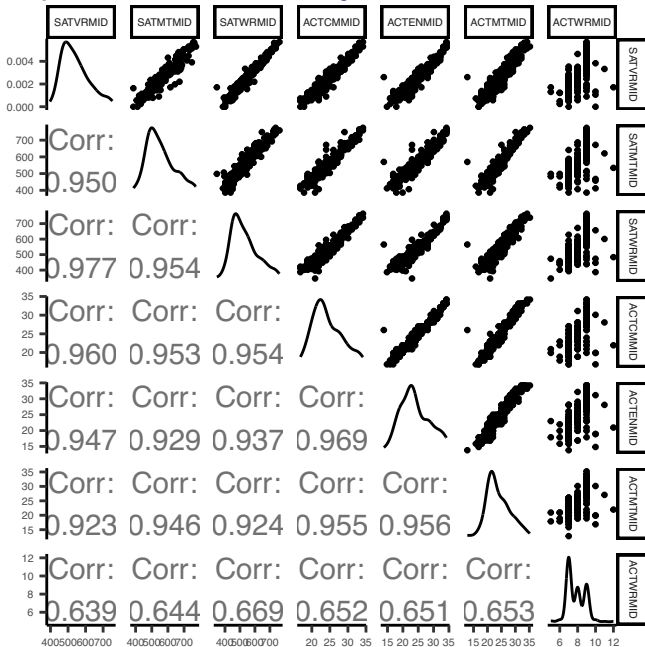
$h_i =$ i^{th} diagonal entry of hat matrix (leverage)

Handling outliers

Should I just remove any unusual observations in the data?

- can remove clear errors
- can remove observation from different population
- otherwise; report results with and without
potential outliers

Example: multicollinearity



Example: multicollinearity

With all the test score variables:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-11.056	1.930	-5.728	0.000
SATVRMID	-0.005	0.011	-0.485	0.628
SATMTMID	0.001	0.008	0.118	0.906
SATWRMID	0.013	0.010	1.209	0.227
ACTCMMID	0.238	0.188	1.266	0.205
ACTENMID	-0.334	0.153	-2.183	0.029
ACTMTMID	0.370	0.154	2.397	0.017
ACTWRMID	0.036	0.203	0.179	0.858

With just one test score variable:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-9.728	1.350	-7.206	0
SATVRMID	0.018	0.003	7.151	0

Variance inflation factors

$$\text{VIF}(\hat{\beta}_j) = \frac{\text{Var}(\hat{\beta}_j \text{ if all variables in model})}{\text{Var}(\hat{\beta}_j \text{ if } X_j \text{ is only explanatory variable})}$$

$$= \frac{1}{1 - R_j^2}$$

$\uparrow R_j^2$ for WLS of X_j on other explanatory variables

$$\text{VIF} \in [1, \infty)$$

$$\text{as } R_j^2 \uparrow, \text{ VIF} \uparrow$$

Common guidelines: concerned if $\text{VIF} \geq 5$ or $\text{VIF} \geq 10$

Handling multicollinearity

```
m1 <- glm(RPY_3YR_70 ~ .,  
          family = binomial, data = scorecard)  
  
car::vif(m1)
```

```
##    CONTROL  SATVRMID  SATMTMID  SATWRMID  ACTCMMID  ACTEN  
##  3.539098 19.586976 12.138599 18.767795 13.158163 12.693  
##      UGDS  PCTFLOAN      NPT4  
##  2.511964  2.124875  2.438207
```

What should we do when there is high multicollinearity?

Summary

- ▶ Outliers:
 - ▶ Change our coefficient estimates
 - ▶ Identify with EDA before model fitting
 - ▶ Also check with Cook's distance after model fitting
 - ▶ Clear errors, or points from a different population of interest, can be removed. Otherwise, report results with and without outliers
- ▶ Multicollinearity:
 - ▶ Makes it hard to interpret meaning of a coefficient (how do we hold other variables constant??)
 - ▶ Increases variability of estimated coefficients
 - ▶ Identify with pairs plots, correlation matrices before model fitting
 - ▶ Also check with VIFs after model fitting
 - ▶ Options: use a subset of the explanatory variables; combine several explanatory variables (average, PCA, etc.); ignore (if only predicting)